Delma Díaz Alcántara

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EMPLOYMENT

Visa International, Mexico city- *Manager Data Engineer – Marketing Team*

Jul DEL 2025-ACTUALIDAD

- **Architected end-to-end, high-throughput data pipelines** and multi-agent workflows using Model Context Protocols (MCPs) and Claude to automate marketing analytics, campaign metrics, and cross-functional enterprise operations. (Based on Docker)
- **Designed and deployed specialized agents** to orchestrate Apache **Airflow** DAG executions, monitor job status, perform automated data quality checks, validate dynamic outputs, optimize the number of tokens and reduce hallucinations risk.
- **Built resilient operational workflows** featuring automated runtime error notifications, proactive alert triage, **incident detection**, and **self-remediation mechanisms.** Integrating Data quality rules using deque.
- **Developed RAG** (Retrieval-Augmented Generation) **pipelines** integrated with Microsoft Copilot agents and Microsoft Sentinel to **streamline text extraction from PowerPoint** pitch decks, significantly reducing executive consulting overhead.
- Programmatically **built dynamic RESTful reporting dashboards** using **Flask and integrated GenAI insights** into Tableau via TabPy and advanced prompt engineering.
- **Implemented agent evaluation and execution harnesses** to optimize agentic reasoning, prompt execution paths, loops, context provisioning, MCP tool-calling, and decision-making accuracy. Improving the performance of agentic workflows.

Tech Stack: Model Context Protocol (MCP), Claude, Microsoft Copilot, Microsoft Sentinel, RAG, Apache Airflow, Python. Agile method.

Core Domains: Agentic AI Systems, RAG Pipelines, Enterprise Automation, Incident Remediation.

Courses: LangChain Antropic, MCPs and Rags

Visa International, Mexico city - *Data Engineer – Core Team*

Apr 2023 - Jul 2025

- Delivered critical engineering solutions and system stability across the entire GLAC (Global Access) ecosystem.

- Engineered, scheduled, and monitored high-throughput ETL/ELT pipelines leveraging Azure Cloud services, Jenkins, and Apache Airflow.
- Architected migration strategies from legacy data workflows to Apache Spark, achieving significant reductions in processing latency and operational costs.
- Established automated data governance and quality control frameworks using AWS Deequ and custom GenAI validation tools integrated with Claude.
- Leveraged Advanced Generative AI tools (ChatGPT, GitHub Copilot, Claude) to accelerate code generation, automate complex debugging, and streamline technical insight extraction.
- Designed system prompts and context-aware agentic workflows using Microsoft Copilot to streamline cross-departmental operations and business decision-making.

Tech Stack: Apache Spark, Azure Cloud, Apache Airflow, Jenkins, Deequ, GenAI (Claude, ChatGPT, GitHub Copilot)**,** Microsoft Copilot Studio, Prompt Engineering, Agentic Workflows.

Banco Azteca, Mexico city- *Data scientist*

Jun 2022-Feb 2023

- Architected end-to-end, AI-driven data pipelines using Azure Cloud and Jenkins to automate complex enterprise workflows and derive strategic business insights from large-scale datasets (Tableau).
- Engineered and deployed production-grade Machine Learning models for regression, clustering (k-means and KNN), feature selection, redes neuronales recurrentes (RNN), and spatial/statistical clustering using PyTorch, TensorFlow, and PySpark to optimize ATM supply chain logistics and analyze client behavior.
- Applied high-performance numerical computing frameworks (NumPy, Pandas) for data indexing, feature engineering, and automated dataset preprocessing.
- Established robust DevOps and MLOps standards by structuring repository architectures, automating CI/CD pipelines, and maintaining technical documentation across GitLab and Jira.

Tech Stack: PySpark, Scala, SQL, Hue, Jira.

Core ML/AI: Time-Series Anomaly Detection, Genetic Algorithms, Supervised & Unsupervised Learning.

NielsenIQ, Mexico city- *Data Analyst*

Mar 2022-Jun 2023

- Designed and deployed automated data quality and anomaly detection pipelines, leveraging machine learning and statistical modeling to identify market trend shifts.
- Developed Python workflows incorporating Genetic Algorithms and Taylor Series expansions for real-time time-series anomaly detection and signal smoothing.
- Engineered predictive reporting pipelines to forecast market dynamics using statistical sampling and probabilistic modeling.

Tech Stack: Python (NumPy, SciPy, Pandas, Keras and Pytorch)

Core Methods: Time-Series Anomaly Detection, Genetic Algorithms, Taylor Series, Statistical Sampling.

GEOEXPLORA, Mexico city-*Project Engineer*

Jul 2019-Jun 2021

- Collected and evaluated field geophysical data, designing solutions using classical statistics and Machine Learning.
- Generated hydrological and seismic isoperiod maps utilizing advanced clustering and QGIS geospatial visualization.
- Led a multidisciplinary team, overseeing field operations, task allocation, and core data analysis pipelines.
- Led the end-to-end preparation of project proposals, statement of work documents, and cost estimations for potential clients.

Tech Stack: Fortran, Linux, Python (NumPy, Pandas, Scikit-Learn, PyTorch), GitHub.

Servicio Sismológico Nacional, Mexico city- *Intern*

- Developed time-series pattern recognition models to analyze embedded signals in complex seismic datasets.
- Processed, classified, and visualized large-scale seismic time-series data using custom Python and geospatial workflows.
- Implemented machine learning algorithms to enhance the detection of volcanic activity from seismic records.

Tech Stack: Python (PyTorch, Scikit-Learn, NumPy, Pandas), Fortran, Linux, Git/GitHub.

Core ML/AI: Deep Learning (CNNs, RNNs for Time-Series Modeling).

EDUCATION

B.S. in Geophysical Engineering & Geotechnics

Aug 2012 -Jul 2017

- **Focus**: Scientific computing in Seismology
- **Thesis**: Pattern estimation preceding volcanic events at Popocatépetl using seismic records.

M.S. in Computer Science and Engineering

Aug 2020 -Jul 2022

- **Focus:** Data Science and Artificial Intelligence Techniques.
- **Thesis:** Classification of spectrograms (seismic record images) using Machine Learning approaches.

ACHIEVEMENTS

Interests & Hobbies: Marathon & Half-Marathon Runner | Earth Sciences & Seismology Enthusiast | Applied Mathematics & Data Science.

SKILLS

Programming Languages

Python (Advanced), Fortran (Advanced), Java (Intermediate), Scala, R, Julia, SQL.

Data Engineering & Cloud

Azure, Apache Spark, PySpark, Airflow, Jenkins, Hadoop, HDFS, Hive

AI & Data Science

Generative AI (GenAI), Copilot Agents, PyTorch, Scikit-Learn, Pandas, NumPy, Mahout, Machine learning, claude

Data Quality & DevOps

Deequ, MCPs, Microsoft Sentinel, Git, GitHub, GitLab, Jira.

Geospatial & Visualization

QGIS, Tableau, AutoCAD (Advanced).

Languages

Spanish (Native), English (Advanced - C1).